03063548328 Sonu Ilyas Mughal

Jan-03-2K21

MTH401 - Differential Equations



Grand Quiz 2020 (Covid-19)

100% Correct Solve (Alhamdulilah)

BY Virtual Assistance Sonu Ilyas Mughal

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Time Left sec(s)

MTH401:Grand Quiz

Quiz Start Time: 03:11 PM, 03 January 202

Question # 1 of 30 (Start time: 03:11:59 PM, 03 January 2021)

Total Marks:

The variation of parameters consists of finding a function $u_1(x)$ such that, $y_p = u_1(x) + y_1(x)$, is a _____ solution of the non - homogeneous differential equation, $\frac{dy}{dx} + P(x)y = f(x)$.

Select the correct option

Reload Math Equations

- complementary
- particular
- general
- exact

Click to Save Answer & Move to Next Question

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MTH401:Grand Quiz

Quiz Start Time: 03:11 P

Question # 2 of 30 (Start time: 03:12:28 PM, 03 January 2021)

Which of the following is true about

$$f(x,y) = x^2 - y^2 + 3?$$

Select the correct option

Reloa

- [x] $f(tx,ty) \neq t^2 f(x,y)$
- [ ] $f(tx,ty) = t^2 f(x,y)$
- [ ] $f(tx,ty) = -t^2 f(x,y)$
- [ ] $f(tx,ty) = tf(x,y)$

Click to Save Answer & Move t

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MTH401:Grand Quiz

Quiz Start Time: 03:11 PM, 03 Jan

Question # 3 of 30 (Start time: 03:12:56 PM, 03 January 2021)

Tota

The differential equation, $\frac{dy}{dx} + y = xy^3$, is an example of _____ differential equation.

Select the correct option

Reload Math Eq

- ○ linear
- ○ exact
- ○ non - exact
- ◉ Bernoulli

Click to Save Answer & Move to Next Qu

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Quiz Start Tim

Question # 4 of 30 (Start time: 03:13:18 PM, 03 January 2021)

The degree of the differential equation: $\frac{dy}{dx} + 5\left(\frac{d^2y}{dx^2}\right)^3 - 3y = e^{\sin x}$ is-----.

Select the correct option

○	0
○	1
○	6
●	3

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MTH401:Grand Quiz

Quiz Start Time: 03:

Question # 5 of 30 (Start time: 03:13:41 PM, 03 January 2021)

A set of functions

$$\{f_1(x), f_2(x), f_3(x), \ldots\ldots, f_n(x)\}$$

is said to be _____ on an interval I if there exist constants

Select the correct option

Re

- ○ linear dependent
- ◉ linear independent

Click to Save Answer & Mo

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MTH401:Grand Quiz

Quiz Start Time: 03:11 PM, 0

Question # 6 of 30 (Start time: 03:14:02 PM, 03 January 2021)

$$The\ differential\ equation\ (1+\ln xy)dx+\left(1+\frac{x}{y}\right)dy=0\ is\ exact\ because\ ---$$

Select the correct option

Reload Ma

- $\frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}=\frac{1}{x}$
- $\frac{\partial M}{\partial x}=\frac{\partial N}{\partial y}=\frac{1}{y}$
- $\frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}=\frac{1}{y}$
- $\frac{\partial M}{\partial x}=\frac{\partial N}{\partial y}=\frac{1}{x}$

Click to Save Answer & Move to Ne

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Question # 7 of 30 (Start time: 03:14:24 PM, 03 January 2021)

In a Bernoulli equation

$$\frac{dy}{dx} - \frac{1}{xy} = xy^3$$

Select the correct option

- $\frac{1}{x}, x\&3$
- $y, xy^3\&3$
- $-y, xy^3\&3$
- $-\frac{1}{x}, x\&3$

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Question # 8 of 30 (Start time: 03:14:44 PM, 03 January 2021)

The periodic time is given by

Select the correct option

○	$\frac{\omega}{2\pi}$
◉	$\frac{2\pi}{\omega}$
○	$2\pi * \omega$
○	$\frac{\pi}{\omega}$

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Quiz Start Time: 03:11 PM, 03 January 2

Question # 9 of 30 (Start time: 03:15:07 PM, 03 January 2021)

Total Mark

Which of the following substitution will transform the differential equation : $\frac{dy}{dx} = \frac{x+y+1}{x+2y+1}$, *in to separable form?*

Select the correct option

Reload Math Equatio

- ○ $y = v + x$
- ○ $y = vx$
- ○ $x = vy$
- ● $x = X + h, y = Y + k$

Click to Save Answer & Move to Next Questio

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Quiz Start Time: 03:

Question # 10 of 30 (Start time: 03:15:28 PM, 03 January 2021)

Which of the following is first order linear equation in unknown variable y?

Select the correct option

- $x\frac{dy}{dx} + (\sin x)\, y = \cos x$
- $y\frac{dx}{dy} + (\sin y)\, x = \cos y$
- $y\frac{dx}{dy} + (\sin y)\, x = \cos x$
- $y\frac{dx}{dy} + (\sin x)\, x = \cos y$

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Quiz Start Time: 03:11 PM,

Question # 11 of 30 (Start time: 03:15:46 PM, 03 January 2021)

What is annihilator operator of the function

$$g(x) = 5$$

?

Select the correct option

Reload M

- [x] D
- [ ] D+1
- [ ] D+5
- [ ] 5D

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Quiz Start Ti

Question # 12 of 30 (Start time: 03:16:06 PM, 03 January 2021)

The gravitational force exerted by the earth on a body of mass m is called ---------- of the body.

Select the correct option

- ○ Force
- ◉ weight

Click to Save Answe

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MTH401:Grand Quiz

Quiz Start Time: 03:11 P

Question # 13 of 30 (Start time: 03:16:26 PM, 03 January 2021)

The root(s) of the auxiliary equation, $am^2 + bm + c = 0$, for the differential equation, $ay'' + by' + cy = 0$, is (are) _____.

Select the correct option

Reloa

○	(I) $m_1 = \dfrac{-b + \sqrt{b^2 - 4ac}}{2a}$
○	(II) $m_2 = \dfrac{-b - \sqrt{b^2 - 4ac}}{2a}$
○	(III) $m_3 = \dfrac{b + \sqrt{b^2 + 4ac}}{2a}$
●	(IV) Both (I) and (II).

Click to Save Answer & Move t

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Question # 14 of 30 (Start time: 03:16:49 PM, 03 January 2021)

Which of the following function would satisfy: $\frac{dy}{dx} = \frac{d^2y}{dx^2} = \cdots = \frac{d^ny}{dx^n}$?

Select the correct option

●	$y = Ae^x$
○	$y = Axe^x$
○	$y = Ae^{x^n}$
○	$y = Ax^n e^{x^n}$

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Question # 15 of 30 (Start time: 03:17:06 PM, 03 January 2021)

For $f(x,y) = x^2 - y^2$, $f(tx,ty) = \ldots$.

Select the correct option

- $f(x,y)$
- $tf(x,y)$
- $t^2 f(x,y)$
- $t^3 f(x,y)$

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Quiz Start Time: 03:11 PM, (

Question # 16 of 30 (Start time: 03:17:26 PM, 03 January 2021)

The annihilator operator for the function, $g(x) = 2e^{-x}$, is _____.

Select the correct option

Reload M

- D
- D^2
- $D+1$
- $D+2$

Click to Save Answer & Move to Ne

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Question # 17 of 30 (Start time: 03:17:48 PM, 03 January 2021)

The differential operator D transforms a _____ function into another function.

Select the correct option

- integrable
- differentiable
- continuous
- discontinuous

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Quiz Start Time: 03:11 PM, 03

Question # 18 of 30 (Start time: 03:18:15 PM, 03 January 2021)

Integrating factor for the first order linear differential equation : $(x-1)^3\frac{dy}{dx}+4(x-1)^2y=\cos\ x$ *is* $---$

Select the correct option

Reload Mat

○	e^{x-1}
■	$(x-1)^4$
○	$\ln(x-1)$
○	$e^{\frac{1}{x-1}}$

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Quiz Start Time: 03:11 PM, 03 Januar

Question # 19 of 30 (Start time: 03:18:40 PM, 03 January 2021)

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Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = 5x^2 \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equa

- $(A_1 x^2 + B_1 x + C_1)\cos 4x + (A_2 x^2 + B_2 x + C_2)\sin 4x$
- $(Ax + B)e^{3x}\cos 4x + (Cx + D)e^{3x}\sin 4x$
- $Ae^{3x}\cos 4x + Be^{3x}\sin 4x$
- $A\cos 4x + B\sin 4x$

Click to Save Answer & Move to Next Ques

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Quiz Start Time: 03:11 P

Question # 20 of 30 (Start time: 03:18:59 PM, 03 January 2021)

The input function $g(x)$ can have one of the forms: A constant function k; A polynomial function; _____

Select the correct option

Reloa

- ○ (I) An exponential function $e^{\alpha x}$
- ○ (II) The trignometric functions $\sin \beta x$, $\cos \beta x$
- ○ (III) Finite sums and products of these functions
- ◉ (IV) All (I), (II) and (III).

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Quiz Start Time: 03:11 P

Question # 21 of 30 (Start time: 03:19:20 PM, 03 January 2021)

Wronskian, $W(x, -3x) =$ _____.

Select the correct option

Reloa

- [x] 0
- [ ] 1
- [ ] $-3x$
- [ ] $3x$

Click to Save Answer & Move t

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Quiz Start Time

Question # 22 of 30 (Start time: 03:19:42 PM, 03 January 2021)

The annihilator operator for the function, $g(x) = 2e^{-x}$, is _____.

Select the correct option

- ○ D
- ◉ D^2
- ○ $D+1$
- ○ $D+5$

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Question # 23 of 30 (Start time: 03:20:07 PM, 03 January 2021)

Which of the following would be a particular solution of the differential equation: $\frac{dy}{dx} = 4$?

Select the correct option

○	$y = 4x + a$
○	$y = ax + 4$
○	$y = ax + a$
◉	$y = 4x + 4$

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Question # 24 of 30 (Start time: 03:20:26 PM, 03 January 2021)

Which of following function will satisfy the differential equation: $\frac{dy}{dx} = -\lambda y$?

Select the correct option

○	$y = ce^{\lambda x}$
○	$y = -ce^{\lambda x}$
◉	$y = ce^{-\lambda x}$
○	$y = -ce^{-\lambda x}$

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Quiz Start Time: 03:11 PM, 03 Janu

Question # 25 of 30 (Start time: 03:20:47 PM, 03 January 2021)

Total

The general linear ordinary differential equation of order n is: $a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_2(x)\frac{d^2 y}{dx^2} + a_1(x)\frac{dy}{dx} + a_0(x)y = 0$, where -------.

Select the correct option

Reload Math Eq

- ○ $a_0(x) = 0$
- ○ $a_0(x) \neq 0$
- ○ $a_n(x) = 0$
- ◉ $a_n(x) \neq 0$

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Quiz Start Time: 03:1

Question # 26 of 30 (Start time: 03:21:10 PM, 03 January 2021)

The velocity of a particle moving with simple harmonic motion is ---------- at the mean position.

Select the correct option

- [x] Zero
- [ ] Maximum
- [ ] Minimum
- [ ] None of these

Click to Save Answer & Mov

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Quiz Start Time: 03:11 PM,

Question # 27 of 30 (Start time: 03:21:31 PM, 03 January 2021)

The auxiliary equation of the differential equation, $3y''' + 5y'' + 10y' - 4y = 0$, is _____.

Select the correct option

Reload M

- $(3m^3 + 5m^2 + 10m - 4)\, e^{mx} = 0$
- $3m^3 + 5m^2 + 10m - 4 = 0$
- $3m^3 + 5m^2 + 10m$
- $3m^3 + 5m^2 - 4$

Click to Save Answer & Move to N

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Quiz Start Time: 03:1

Question # 28 of 30 (Start time: 03:21:54 PM, 03 January 2021)

Vibration of an object on equilibrium point is called simple harmonic motion when the restoring force is proportional to _____.

Select the correct option

- ○ A spring constant
- ◉ Displacement
- ○ Time
- ○ Mass

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Question # 29 of 30 (Start time: 03:22:15 PM, 03 January 2021)

The family of curve

$$x^2 + y^2 = C^2$$

represents a family of circles centered at _____.

Select the correct option

○	(1,0)
○	(0,1)
◉	(0,0)
○	(1,1)

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Question # 30 of 30 (Start time: 03:22:34 PM, 03 January 2021)

A set of functions

$$\{f_1(x), f_2(x), f_3(x), \ldots\ldots, f_n(x)\}$$

is said to be _____ on an interval I if there exist constants

Select the correct option

- [x] linear dependent
- [ ] linear independent

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Question # 1 of 30 (Start time: 03:28:21 PM, 03 January 2021)

What is annihilator operator of the function

$$g(x) = 5$$

?

Select the correct option

◉	D
○	D+1
○	D+5
○	5D

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Question # 2 of 30 (Start time: 03:28:43 PM, 03 January 2021)

$\frac{d^2y}{dx^2} + 5\left(\frac{dy}{dx}\right)^3 - 3y = e^{\sin x}$ is an example of -------- differential equation.

Select the correct option

○	ordinary linear
◉	ordinary non-linear
○	partial linear
○	partial non-linear

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Question # 3 of 30 (Start time: 03:29:04 PM, 03 January 2021)

If

$$y = 2 + x$$

, then which of the following is true for it?

Select the correct option

○	Its annihilator operator is D
●	Its annihilator operator is D^2
○	Its annihilator operator is D^3
○	Its annihilator operator is D+1

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Quiz S

Question # 4 of 30 (Start time: 03:29:25 PM, 03 January 2021)

Which of following is the Homogeneous function?

Select the correct option

●	$f(x,y)=\frac{x}{x+y}$
○	$f(x,y)=\frac{1}{x+y}$
○	$f(x,y)=\frac{x}{1+y}$
○	$f(x,y)=\frac{1}{1+y}$

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Quiz Start Ti

Question # 5 of 30 (Start time: 03:29:43 PM, 03 January 2021)

The gravitational force exerted by the earth on a body of mass m is called ---------- of the body.

Select the correct option

- ○ Force
- ◉ weight

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Quiz Start Time: 03:28 PM

Question # 6 of 30 (Start time: 03:30:01 PM, 03 January 2021)

The differential equation of orthogonal trajectory to the family of curves

$$x^2 + y^2 = C^2$$

is _____.

Select the correct option

Reload

- $\frac{dy}{dx} = \frac{x}{y}$
- $\frac{dy}{dx} = \frac{y}{x}$
- $\frac{dy}{dx} = -\frac{x}{y}$
- $\frac{dy}{dx} = -\frac{y}{x}$

Click to Save Answer & Move to

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Quiz Start Time: 03:2

Question # 7 of 30 (Start time: 03:30:22 PM, 03 January 2021)

The annihilator operator for the function, $g(x) = 2e^{-x}$, is _____.

Select the correct option

Re

- D
- D^2
- $D+1$
- $D+2$

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MTH401:Grand Quiz

Quiz Start Time

Question # 8 of 30 (Start time: 03:30:44 PM, 03 January 2021)

The annihilator operator for the function, $g(x) = 5x$, is _____.

Select the correct option

○	D
●	D^2
○	$D+1$
○	$D+5$

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Quiz Start Time

Question # 9 of 30 (Start time: 03:31:03 PM, 03 January 2021)

If 3,3 are real roots of a differential equation, then the general solution is_____.

Select the correct option

○	$y = c_1 e^{3x} + c_2 e^{3x}$
○	$y = c_1 e^{3x} - c_2 e^{3x}$
●	$y = (c_1 + c_2 x) e^{3x}$
○	$y = c_1 e^{3x}$

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Quiz Start Time: 03:28 PM, 03 Jan

Question # 10 of 30 (Start time: 03:31:24 PM, 03 January 2021) Tot

If we want to solve an initial value problem then we need to to apply the initial conditions to the _____ solution and not to the complementary function alone.

Select the correct option

- complementary
- particular
- general
- exact

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Quiz S

Question # 11 of 30 (Start time: 03:31:45 PM, 03 January 2021)

Which of following function will satisfy the differential equation: $\frac{dy}{dx} = -\lambda y$?

Select the correct option

○	$y = ce^{\lambda x}$
○	$y = -ce^{\lambda x}$
●	$y = ce^{-\lambda x}$
○	$y = -ce^{-\lambda x}$

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Quiz Start Time: 03:28 PM, 03 January

Question # 12 of 30 (Start time: 03:32:05 PM, 03 January 2021)

Total Ma

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = \cos 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equati

○	Ae^{4x}
○	$Ax^2 + Bx + C$
●	$A\cos 4x + B\sin 4x$
○	$(A + B)(\cos 4x + \sin 4x)$

Click to Save Answer & Move to Next Questi

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Quiz Start Ti

Question # 13 of 30 (Start time: 03:32:30 PM, 03 January 2021)

The differential equation

$$4x^2\frac{d^2y}{dx^2} + 12x\frac{dy}{dx} + 3y = 0$$

Select the correct option

- (●) Initial value problem
- ( ) Boundary value problem
- ( ) a & b both
- ( ) None of these

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MTH401:Grand Quiz

Quiz Start T

Question # 14 of 30 (Start time: 03:32:52 PM, 03 January 2021)

The differential equation

$$\frac{dy}{dx} - y = y^3$$

Select the correct option

●	Bernoulli's
○	Homogeneous
○	Cauchy
○	Bessel

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Question # 15 of 30 (Start time: 03:33:12 PM, 03 January 2021)

What is annihilator operator of the function

$$g(x) = 4\sin x$$

?

Select the correct option

○	$(D^2 - 1)$
○	$(D^2 + 4)$
○	$(D^2 - 4)$
◉	$(D^2 + 1)$

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Question # 16 of 30 (Start time: 03:33:33 PM, 03 January 2021)

In differential equations, the symbol D is known as _____ operator.

Select the correct option

- [x] differential
- [ ] arithmetic
- [ ] relational
- [ ] integral

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Quiz Star

Question # 17 of 30 (Start time: 03:33:58 PM, 03 January 2021)

In exponential model for the population growth

$$P(t) = P_0 e^{kt} \quad If\ k > 0,\ then\ \lim_{t\to\infty} P(t) =$$

Select the correct option

○	0
○	1
●	∞
○	- ∞

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Quiz Start Time: 03:28 PM, 03 Ja

Question # 18 of 30 (Start time: 03:34:19 PM, 03 January 2021)

To

If the auxiliary equation, $am^2 + bm + c = 0$, has two unequal real roots m_1 and m_2 then the solutions of the differential equation, $ay'' + by' + cy = 0$, could be _____.

Select the correct option

Reload Math E

- (I) $y = c\,e^{mx}$
- (II) $y_1 = e^{m_1 x}$
- (III) $y_2 = e^{m_2 x}$
- (IV) Both (II) and (III).

Click to Save Answer & Move to Next Q

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Quiz Start Time: 0

Question # 19 of 30 (Start time: 03:34:39 PM, 03 January 2021)

Which of the following is an equivalent form of the exact differential equation:
$ydx + xdy = 0$?

Select the correct option

○	$d(\frac{x}{y}) = 0$
○	$d\left(\frac{y}{x}\right) = 0$
●	$d(xy) = 0$
○	$d(x + y) = 0$

Click to Save Answer & M

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Quiz Start Time: 03:28 PM

Question # 20 of 30 (Start time: 03:35:01 PM, 03 January 2021)

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$. *If* the input function, $g(x) = (9x - 2)e^{5x}$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload

- Ae^{5x}
- $(Ax + B)e^{5x}$
- $(Ax^2 + Bx + C)e^{5x}$
- $Ae^{5x} \cos 5x + Be^{5x} \sin 5x$

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Quiz Start Time: 03:28 PM, 03 January 2

Question # 21 of 30 (Start time: 03:35:26 PM, 03 January 2021)

Total Mark

Which of following is the integrating factor for the 1st *order linear differential equation :* $\frac{dy}{dx} + y = f(x)$

Select the correct option

Reload Math Equatio

○	$\ln x$
○	x
○	$\frac{1}{x}$
●	e^x

Click to Save Answer & Move to Next Questio

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MTH401:Grand Quiz

Quiz Start Time:

Question # 22 of 30 (Start time: 03:35:50 PM, 03 January 2021)

Wronskian, $W(x, -3x) =$ _____.

Select the correct option

- [x] 0
- [ ] 1
- [ ] $-3x$
- [ ] $3x$

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MTH401:Grand Quiz Quiz

Question # 23 of 30 (Start time: 03:36:09 PM, 03 January 2021)

If

$$y = e^{5x}$$

, then which of the following is true for it?

Select the correct option

○	Its annihilator operator is (D+5)
●	Its annihilator operator is (D-5)
○	Its annihilator operator is $(D+5)^2$
○	Its annihilator operator is $(D-5)^2$

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MTH401:Grand Quiz

Quiz Start Time: (

Question # 24 of 30 (Start time: 03:36:28 PM, 03 January 2021)

The differential operator L is called an annihilator operator of the function f if: _____.

Select the correct option

- (i) L is a linear differential operator with constant coefficients.
- (ii) y = f(x) defines a sufficiently differentiable function.
- (iii) The function f is such that L(y)=0.
- (iv) All (i), (ii) and (iii).

Click to Save Answer & N

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MTH401:Grand Quiz

Quiz S

Question # 25 of 30 (Start time: 03:36:46 PM, 03 January 2021)

For $f(x,y) = x^2 - y^2$, $f(tx,ty) = \ldots$.

Select the correct option

- $f(x,y)$
- $tf(x,y)$
- $t^2 f(x,y)$
- $t^3 f(x,y)$

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MTH401:Grand Quiz

Quiz Start T

Question # 26 of 30 (Start time: 03:37:07 PM, 03 January 2021)

Vibration of an object on equilibrium point is called simple harmonic motion when the restoring force is proportional to _____.

Select the correct option

- A spring constant
- Displacement
- Time
- Mass

Click to Save Answ

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MTH401:Grand Quiz

Quiz Start Time: 03:28 PM, 0

Question # 27 of 30 (Start time: 03:37:28 PM, 03 January 2021)

The integrating factor for the first order linear differential equation : $\frac{dy}{dx} + y \cot x = \sin^2 x$ *is* $- - -$

Select the correct option

Reload Ma

- [x] $\sin x$
- [ ] $\cos x$
- [ ] $e^{\sin x}$
- [ ] $e^{\cos x}$

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Quiz Start Time: 03:28

Question # 28 of 30 (Start time: 03:37:50 PM, 03 January 2021)

$$For\ f(x,y) = \frac{2x}{3y} + 7, f(tx,ty) = - - -$$

Select the correct option

Relo

- [x] $f(x,y)$
- [ ] $tf(x,y)$
- [ ] $tf(x,y)$
- [ ] $t^3 f(x,y)$

Click to Save Answer & Move

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MTH401:Grand Quiz

Quiz Start Time: 03:28 PM, 03

Question # 29 of 30 (Start time: 03:38:11 PM, 03 January 2021)

The integrating factor for the first order linear differential equation : $\frac{dy}{dx} + \frac{y}{x \ln x} = x^2$ *is* $---$.

Select the correct option

Reload Mat

- e^{e^x}
- e^x
- $\ln|\ln x|$
- $\ln x$

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MTH401:Grand Quiz

Quiz Start Time: (

Question # 30 of 30 (Start time: 03:38:31 PM, 03 January 2021)

The displacement measured below the equilibrium position are _____.

Select the correct option

- (●) positive
- ( ) negative

Click to Save Answer & N

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MTH401:Grand Quiz

Quiz Start Time: 03:4

Question # 1 of 30 (Start time: 03:41:45 PM, 03 January 2021)

The superposition principle is a property of linear differential equations and it does not hold in case of _____ differential equations.

Select the correct option

- linear
- nonlinear

Click to Save Answer & Mov

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MTH401:Grand Quiz

Quiz Start Time: 03:41 PM, 03 Ja

Question # 3 of 30 (Start time: 03:43:04 PM, 03 January 2021) Tc

Wronskian

$$W(1, e^x)$$

=_____.

Select the correct option

Reload Math

○	0
○	1
○	-1
◉	e^x

Click to Save Answer & Move to Next Q

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MTH401:Grand Quiz

Quiz Start Time: 03:4

Question # 4 of 30 (Start time: 03:43:28 PM, 03 January 2021)

Classify the following differential equation

$$e^x \frac{dy}{dx} + 2y = 3xy$$

Select the correct option

Re

○	Separable and not linear
○	Linear and not separable
●	Both separable and linear
○	Neither separable nor linear

Click to Save Answer & Mov

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MTH401:Grand Quiz

Quiz Start Time:

Question # 5 of 30 (Start time: 03:43:55 PM, 03 January 2021)

If initial amount of a radioactive isotope is 100g. What will be the amount at the end of 30 days such that K=0.043?

Select the correct option

- ○ 371.415
- ○ 380.560
- ◉ 363.279
- ○ 360.351

Click to Save Answer &

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Quiz Start Time: 03:41 PM, 03

Question # 6 of 30 (Start time: 03:44:24 PM, 03 January 2021)

In the population growth model, the solution of the differential equation is given by _____.

Select the correct option

Reload Math

○	$A(t) = ke^{kt}$
●	$A(t) = A_0 e^{kt}$
○	$A(t) = A_0/2$
○	$A(t) = A_0/2$

Click to Save Answer & Move to Next

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Quiz Start Time: 03:41 PM, 03 Jar

Question # 7 of 30 (Start time: 03:44:53 PM, 03 January 2021)

Tot

$If\ d\left[\ln(xy)\right] = 0\ is\ anequivalent\ form\ of\ the\ differential\ equation : xdy + ydx = 0, then\ its\ solutionis\ - - -$

Select the correct option

Reload Math E

○	$y = c + x$
○	$y = c - x$
○	$y = cx$
●	$y = \frac{c}{x}$

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Quiz Start Time: 03:41 PM, 03 J

Question # 8 of 30 (Start time: 03:46:44 PM, 03 January 2021)

In a Bernoulli equation

$$\frac{dy}{dx} - \frac{1}{xy} = xy^3$$

Select the correct option

Reload Math

- $\frac{1}{x}, x\&3$
- $y, xy^3\&3$
- $-y, xy^3\&3$
- $-\frac{1}{x}, x\&3$

Click to Save Answer & Move to Next

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Quiz Start Time: 0

Question # 9 of 30 (Start time: 03:47:18 PM, 03 January 2021)

$For f(x,y) = x^2y - xy^2, f(tx,ty) = \ldots$

Select the correct option

○	$f(x,y)$
○	$tf(x,y)$
○	$t^2f(x,y)$
●	$t^3f(x,y)$

Click to Save Answer & M

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Quiz Start Time: 03:41 PM,

Question # 10 of 30 (Start time: 03:47:45 PM, 03 January 2021)

If T be the half - life then solution of the population growth model could be _____.

Select the correct option

Reload

- $\frac{A_0}{2}$
- $A_0\, e^{kt}$
- $\frac{A_0}{2} = A_0\, e^{kt}$
- $\frac{A_0}{2}\left(A_0\, e^{kt}\right)$

Click to Save Answer & Move to N

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Quiz Start Time: 03:

Question # 11 of 30 (Start time: 03:48:05 PM, 03 January 2021)

For a system in simple harmonic motion which of the following is the time required to complete a cycle of motion?

Select the correct option

- ○ Frequency
- ○ Amplitude
- ◉ Period
- ○ Revolution

Click to Save Answer & Mo

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Question # 12 of 30 (Start time: 03:48:28 PM, 03 January 2021)

The differential equation

$$y(1+2xy)dx + x(1-2xy)dy = 0$$

is

Select the correct option

○	Separable
○	Homogeneous
○	Exact
●	None of these

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Quiz Start Time: 03:41 PM,

Question # 13 of 30 (Start time: 03:48:49 PM, 03 January 2021)

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = x^2 e^{5x}$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload

- Ae^{5x}
- $(Ax + B)e^{5x}$
- $(Ax^2 + Bx + C)e^{5x}$
- $Ae^{5x} \cos 5x + Be^{5x} \sin 5x$

Click to Save Answer & Move to N

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Quiz S

Question # 14 of 30 (Start time: 03:49:11 PM, 03 January 2021)

Which of following function will satisfy the differential equation: $\frac{dy}{dx} = -\lambda y$?

Select the correct option

○	$y = ce^{\lambda x}$
○	$y = -ce^{\lambda x}$
●	$y = ce^{-\lambda x}$
○	$y = -ce^{-\lambda x}$

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Quiz S

Question # 15 of 30 (Start time: 03:49:36 PM, 03 January 2021)

The velocity of a particle moving with simple harmonic motion is ---------- at the mean position.

Select the correct option

- (●) Zero
- ( ) Maximum
- ( ) Minimum
- ( ) None of these

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Quiz Start Time: 03:41 P

Question # 16 of 30 (Start time: 03:49:55 PM, 03 January 2021)

Vibration of an object on equilibrium point is called simple harmonic motion when the restoring force is proportional to _____.

Select the correct option

- ○ A spring constant
- ◉ Displacement
- ○ Time
- ○ Mass

Click to Save Answer & Move t

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Quiz Start Time: 03:41 PM,

Question # 17 of 30 (Start time: 03:50:15 PM, 03 January 2021)

If the general solution of a separable differential equation is $\sin^{-1} y = \cos^{-1} x + c$, provided that $y(\frac{1}{\sqrt{2}}) = \frac{1}{\sqrt{2}}$, then $c = - - - -$.

Select the correct option

Reload

- ○ $\frac{\pi}{2}$
- ○ $\frac{\pi}{4}$
- ○ $-\frac{\pi}{2}$
- ◉ 0

Click to Save Answer & Move to N

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Question # 18 of 30 (Start time: 03:50:36 PM, 03 January 2021)

In differential equations, the symbol D is known as _____ operator.

Select the correct option

- [x] differential
- [ ] arithmetic
- [ ] relational
- [ ] integral

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Question # 19 of 30 (Start time: 03:50:59 PM, 03 January 2021)

Which of the following is an example of partial differential equation?

Select the correct option

○	$(y-x)\,dx+4xdy=0$
●	$\frac{\partial u}{\partial y}=-\frac{\partial v}{\partial x}$
○	$ax^2+bx+c=0, a\neq 0$
○	$\varphi(x)=f(x)+\lambda\int_a^b \kappa(x,t)\varphi(t)dt$

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Quiz Start Time: 03:41 PM, (

Question # 20 of 30 (Start time: 03:51:21 PM, 03 January 2021)

The logistic equation, $\frac{dP}{dt} = P(a - bP)$, can be identified as a nonlinear equation that is _____.

Select the correct option

Reload M

- integrable
- separable

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Quiz Start Time: 03:41 PM, 03 J

Question # 21 of 30 (Start time: 03:51:41 PM, 03 January 2021)

T

On comparing the differential equation:dy = dx with standard form of M(x,y)dx + N(x,y)dy = 0, we obtain - - - -

Select the correct option

Reload Math

- $M(x,y) = N(x,y)$
- $|M(x,y)| = |N(x,y)|$
- $\frac{\partial M(x,y)}{\partial x} \neq \frac{\partial N(x,y)}{\partial y}$
- $\frac{\partial M(x,y)}{\partial y} \neq \frac{\partial N(x,y)}{\partial x}$

Click to Save Answer & Move to Next

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Quiz Start Time: 03

Question # 22 of 30 (Start time: 03:52:01 PM, 03 January 2021)

For a second order linear ODE, if y_1 and y_2 are its solutions, and further $W(y_1, y_2) = 2$, $W_1 = \frac{e^{-x}}{x}$, then $u'_1 =$ _____.

Select the correct option

- [x] $\frac{e^{-x}}{2x}$
- [ ] $-\frac{e^{-x}}{2x}$
- [ ] $\frac{e^{-x}}{x}$
- [ ] $2x\,e^{-x}$

Click to Save Answer & M

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Quiz Start Time: 03:41 PM, 03 Januar

Question # 22 of 30 (Start time: 03:52:01 PM, 03 January 2021)

Total M

For a second order linear ODE, if y_1 and y_2 are its solutions, and further $W(y_1, y_2) = 2$, $W_1 = \dfrac{e^{-x}}{x}$, then $u'_1 =$ _____.

Select the correct option

Reload Math Equa

- [x] $\dfrac{e^{-x}}{2x}$
- [ ] $-\dfrac{e^{-x}}{2x}$
- [ ] $\dfrac{e^{-x}}{x}$
- [ ] $2x\,e^{-x}$

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Quiz Start Time: 03:41

Question # 23 of 30 (Start time: 03:52:40 PM, 03 January 2021)

A set of n functions

$$\{f_1(x), f_2(x), f_3(x), \ldots\ldots, f_n(x)\}$$

is _____ if at least one of them can be expressed as a linear combination of the remaining.

Select the correct option

Reloa

- (●) linear dependent
- ( ) linear independent

Click to Save Answer & Move

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MTH401:Grand Quiz

Question # 24 of 30 (Start time: 03:53:01 PM, 03 January 2021)

The order of the differential equation: $\frac{d^2y}{dx^2} + 5\left(\frac{dy}{dx}\right)^3 - 3y = e^{\sin x}$ is-----.

Select the correct option

○	0
○	1
◉	2
○	3

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MTH401:Grand Quiz

Quiz Start Time: 03:41 PM, 03 January

Question # 25 of 30 (Start time: 03:53:38 PM, 03 January 2021)

Total Ma

Which of following is the integrating factor for the 1st *order linear differential equation :* $\frac{dy}{dx} + y = f(x)$

Select the correct option

Reload Math Equati

○	$\ln x$
○	x
○	$\frac{1}{x}$
●	e^x

Click to Save Answer & Move to Next Questi

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Quiz Start Time: 03:41 PM, 03 January

Question # 25 of 30 (Start time: 03:53:38 PM, 03 January 2021)

Total Ma

Which of following is the integrating factor for the 1st *order linear differential equation :* $\frac{dy}{dx} + y = f(x)$

Select the correct option

Reload Math Equati

○	$\ln x$
○	x
○	$\frac{1}{x}$
●	e^x

Click to Save Answer & Move to Next Questi

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MTH401:Grand Quiz

Question # 26 of 30 (Start time: 03:54:38 PM, 03 January 2021)

Wronskian

$$W(x, 2x)$$

=_____.

Select the correct option

●	0
○	1
○	-1
○	None of these

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Quiz Start Time: 03:41 PM, 03 Ja

Question # 27 of 30 (Start time: 03:55:02 PM, 03 January 2021)

To

$If\ d(\frac{x^2}{y}) = 0\ is\ an\ equivalent\ form\ of\ the\ exact\ differential\ equation: 2xydx - x^2dy = 0, then\ its\ solution\ is\ ---.$

Select the correct option

Reload Math E

- $y = cx$
- $x = cy$
- $x^2 = cy$
- $y = cx^2$

Click to Save Answer & Move to Next Q

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Quiz Start Time: 03:41 P

Question # 28 of 30 (Start time: 03:55:26 PM, 03 January 2021)

$$For\ f(x,y) = \frac{2x}{3y} + 7, f(tx,ty) = - - -$$

Select the correct option

Reload

●	$f(x,y)$
○	$tf(x,y)$
○	$tf(x,y)$
○	$t^3 f(x,y)$

Click to Save Answer & Move to

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Quiz Start Time: 03:41 PM, 03 J

Question # 29 of 30 (Start time: 03:55:50 PM, 03 January 2021)

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math

- Ae^{4x}
- $Ax^2 + Bx + C$
- $A\cos 4x + B\sin 4x$
- $(A + B)(\cos 4x + \sin 4x)$

Click to Save Answer & Move to Next

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MTH401:Grand Quiz

Quiz

Question # 30 of 30 (Start time: 03:56:17 PM, 03 January 2021)

The differential operator D possesses the _____ property.

Select the correct option

- closure
- inverse
- linearity
- nonlinearity

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MTH401:Grand Quiz

Quiz Start Time: 03:58 PM,

Question # 1 of 30 (Start time: 03:58:20 PM, 03 January 2021)

Integrating factor for the first order linear differential equation : $(x-1)^3\frac{dy}{dx}+4(x-1)^2y=\cos\ x$ *is* $---$

Select the correct option

Reload M

- e^{x-1}
- $(x-1)^4$
- $\ln(x-1)$
- $e^{\frac{1}{x-1}}$

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Question # 2 of 30 (Start time: 03:58:53 PM, 03 January 2021)

Which of the following is a boundary value problem(BVP)?

Select the correct option

- $\frac{d^2y}{dx^2} + y = 0$
- $\frac{d^2y}{dx^2} + y = 0, x \in [-2, 2]$
- $\frac{d^2y}{dx^2} + y = 0, y(1) = -2$ and $y'(1) = 2$
- $\frac{d^2y}{dx^2} + y = 0, y(0) = 0$ and $y\left(\frac{\pi}{2}\right) = 2$

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Quiz Start Time: 03:58 PM, (

Question # 3 of 30 (Start time: 03:59:14 PM, 03 January 2021)

For the nonexact differential equation

$$y\left(xy+2x^2y^2\right)dx+x\left(xy-x^2y^2\right)dy=0, if xM(x,y)-yN(x,y)=x^3y^3\neq 0,$$

then the integrating factor is - - -

Select the correct option

Reload M

- $e^{x^3y^3}$
- $e^{-x^3y^3}$
- $\frac{1}{x^3y^3}$
- $\frac{-1}{x^3y^3}$

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MTH401:Grand Quiz

Quiz Start

Question # 4 of 30 (Start time: 03:59:48 PM, 03 January 2021)

Which of the following is an example of ordinary differential equation?

Select the correct option

- [x] $(y-x)\,dx+4xdy=0$
- [ ] $\frac{\partial u}{\partial y}=-\frac{\partial v}{\partial x}$
- [ ] $ax^2+bx+c=0, a\neq 0$
- [ ] $\varphi(x)=f(x)+\lambda\int_a^b \kappa(x,t)\varphi(t)dt$

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MTH401:Grand Quiz

Quiz Start Time: 03

Question # 5 of 30 (Start time: 04:00:13 PM, 03 January 2021)

The differential equation of orthogonal trajectory to the family of curves

$$x^2 + y^2 = C^2$$

is _____.

Select the correct option

○	$\frac{dy}{dx} = \frac{x}{y}$
●	$\frac{dy}{dx} = \frac{y}{x}$
○	$\frac{dy}{dx} = -\frac{x}{y}$
○	$\frac{dy}{dx} = -\frac{y}{x}$

Click to Save Answer & M

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Quiz Start Time: 03:58 PM, 03 Jan

Question # 6 of 30 (Start time: 04:00:39 PM, 03 January 2021)

Tota

If X is the remaining amount of the substance A at any time t then, $\frac{dX}{dt} = kX$, where $k < 0$ because X is _____.

Select the correct option

Reload Math E

- (●) decreasing
- ( ) increasing

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Quiz Star

Question # 7 of 30 (Start time: 04:01:01 PM, 03 January 2021)

Orthogonal trajectories occur naturally in the area(s) of_____.

Select the correct option

- (i) physics
- (ii) fluid dynamics
- (iii) electricity and magnetism
- (iv) All (i), (ii) and (iii).

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MTH401:Grand Quiz

Quiz Start Time: 03:58

Question # 8 of 30 (Start time: 04:01:27 PM, 03 January 2021)

Which of following is the Homogeneous function?

Select the correct option

Relo

- [x] $f(x,y)=\frac{x}{x+y}$
- [ ] $f(x,y)=\frac{1}{x+y}$
- [ ] $f(x,y)=\frac{x}{1+y}$
- [ ] $f(x,y)=\frac{1}{1+y}$

Click to Save Answer & Move

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Question # 9 of 30 (Start time: 04:01:49 PM, 03 January 2021)

The differential operator D possesses the _____ property.

Select the correct option

- closure
- inverse
- linearity
- nonlinearity

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Quiz Start Time: 03:58 PM, 03

Question # 10 of 30 (Start time: 04:02:08 PM, 03 January 2021)

On comparing the differential equation:dy = dx with standard form of M(x,y)dx + N(x,y)dy = 0, we obtain - - - -

Select the correct option

Reload Mat

- ○ $M(x,y) = N(x,y)$
- ◉ $|M(x,y)| = |N(x,y)|$
- ○ $\frac{\partial M(x,y)}{\partial x} \neq \frac{\partial N(x,y)}{\partial y}$
- ○ $\frac{\partial M(x,y)}{\partial y} \neq \frac{\partial N(x,y)}{\partial x}$

Click to Save Answer & Move to Nex

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Quiz Start Tim

Question # 11 of 30 (Start time: 04:02:28 PM, 03 January 2021)

Separable form $f(y)dy + g(x)dx = 0$, of the differential equation: $x \sin y dx + \left(x^2 + 1\right) \cos y dy = 0$ is-----.

Select the correct option

○	$\tan y dy + \frac{x}{x^2+1} dx = 0$
●	$\cot y dy + \frac{x}{x^2+1} dx = 0$
○	$\tan y dy + \frac{x}{x^2-1} dx = 0$
○	$\cot y dy + \frac{x}{x^2-1} dx = 0$

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MTH401:Grand Quiz

Quiz Start Time: 03:58 PM, 03 January 202

Question # 12 of 30 (Start time: 04:02:55 PM, 03 January 2021)

Total Marks:

Which of the following substitution will transform the differential equation : $\frac{dy}{dx} = \frac{x+y+1}{x+y-1}$, *in to separable form?*

Select the correct option

Reload Math Equations

- ○ $x = X + h, y = Y + k$
- ○ $x = vy$
- ○ $y = vx$
- ● $z = x + y$

Click to Save Answer & Move to Next Question

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MTH401:Grand Quiz

Quiz Start Time: 03

Question # 13 of 30 (Start time: 04:03:19 PM, 03 January 2021)

General solution of the separable differential equation: $\frac{\sec^2 y}{\tan y}dy = dx$ is-----.

Select the correct option

○	$y = \tan^{-1}(c + e^x)$
○	$y = \cot(c + e^x)$
●	$y = \tan^{-1}(ce^x)$
○	$y = \cot(ce^x)$

Click to Save Answer & M

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MTH401:Grand Quiz

Quiz Start Time: 03:5

Question # 14 of 30 (Start time: 04:03:44 PM, 03 January 2021)

The differential equation

$$\frac{du}{dx} + \frac{2}{x}dx = 3 - \frac{6}{x}$$

Select the correct option

Re

○	$\frac{1}{x}$
●	x^2
○	$\frac{1}{x^2}$
○	$-\frac{1}{x}$

Click to Save Answer & Mov

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MTH401:Grand Quiz

Quiz Start Time: 03:5

Question # 14 of 30 (Start time: 04:03:44 PM, 03 January 2021)

The differential equation

$$\frac{du}{dx} + \frac{2}{x}dx = 3 - \frac{6}{x}$$

Select the correct option

Re

○	$\frac{1}{x}$
●	x^2
○	$\frac{1}{x^2}$
○	$-\frac{1}{x}$

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Question # 15 of 30 (Start time: 04:04:05 PM, 03 January 2021)

For $f(x,y) = x^2 - y^2$, $f(tx,ty) = \ldots$

Select the correct option

- $f(x,y)$
- $tf(x,y)$
- $t^2 f(x,y)$
- $t^3 f(x,y)$

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MTH401:Grand Quiz

Quiz Start Time: 03:58 PM, C

Question # 16 of 30 (Start time: 04:04:31 PM, 03 January 2021)

Which of following is the integrating factor for the 1st *order linear differential equation :* $\frac{dy}{dx} + y = f(x)$

Select the correct option

Reload Ma

- ○ $\ln x$
- ○ x
- ○ $\frac{1}{x}$
- ◉ e^x

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MTH401:Grand Quiz

Quiz Start Time: 03:58 PM, 03 January

Question # 17 of 30 (Start time: 04:04:55 PM, 03 January 2021)

Total Ma

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = xe^{3x} \cos 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equati

- $(A_1 x^2 + B_1 x + C_1) \cos 4x + (A_2 x^2 + B_2 x + C_2) \sin 4x$
- $(Ax + B)e^{3x} \cos 4x + (Cx + D)e^{3x} \sin 4x$
- $Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
- $A \cos 4x + B \sin 4x$

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MTH401:Grand Quiz

Quiz Start T

Question # 18 of 30 (Start time: 04:05:15 PM, 03 January 2021)

$\frac{d^3y}{dx^3} + y^2 = 0$ is a ------differential equation of degree----.

Select the correct option

○	linear, 1
○	linear, 3
◉	non-linear, 1
○	non-linear, 3

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MTH401:Grand Quiz

Quiz Start Time: 03:58 PM

Question # 19 of 30 (Start time: 04:05:34 PM, 03 January 2021)

The roots of the auxiliary equation, $m^2 - 10m + 25 = (m-5)^2 = 0$, for a differential equation is _____.

Select the correct option

Reload

- $m_1 = m_2 = 25$
- $m_1 = m_2 = 5$
- $m = 25$
- $m = -10$

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MTH401:Grand Quiz

Quiz Start Tin

Question # 20 of 30 (Start time: 04:05:53 PM, 03 January 2021)

A set of n functions

$$\{f_1(x), f_2(x), f_3(x), \ldots\ldots, f_n(x)\}$$

is _____ if at least one of them can be expressed as a linear combination of the remaining.

Select the correct option

- (●) linear dependent
- ( ) linear independent

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MTH401:Grand Quiz

Quiz Start Time: 03:58 PM, 03 Ja

Question # 21 of 30 (Start time: 04:06:18 PM, 03 January 2021)

To

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = 5x^2 \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math E

●	$(A_1x^2 + B_1x + C_1)\cos 4x + (A_2x^2 + B_2x + C_2)\sin 4x$
○	$(Ax + B)e^{3x}\cos 4x + (Cx + D)e^{3x}\sin 4x$
○	$Ae^{3x}\cos 4x + Be^{3x}\sin 4x$
○	$A\cos 4x + B\sin 4x$

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MTH401:Grand Quiz Quiz

Question # 22 of 30 (Start time: 04:06:39 PM, 03 January 2021)

The equation of free un-damped motion is _____.

Select the correct option

- $\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$
- $\frac{d^2x}{dt^2} - \frac{k}{m}x = 0$

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Quiz Start Ti

Question # 23 of 30 (Start time: 04:07:11 PM, 03 January 2021)

In differential equations, the symbol D is known as _____ operator.

Select the correct option

- [x] differential
- [ ] arithmetic
- [ ] relational
- [ ] integral

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MTH401:Grand Quiz

Quiz Start Time: 03:58 PM, 0

Question # 24 of 30 (Start time: 04:07:29 PM, 03 January 2021)

To reduce Bernoulli equation, $\frac{dy}{dx} - y = y^3$, into linear we put $v =$ _____.

Select the correct option

Reload Ma

- y^{-1}
- y^{-2}
- y^{-3}
- y^2

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MTH401:Grand Quiz

Quiz Start Time: 03:58 PM, C

Question # 25 of 30 (Start time: 04:07:48 PM, 03 January 2021)

Separable form $g(x)dx + f(y)dy = 0$, of the differential equation: $(2x\cos y)\,dx + x^2\,(\sec y - \sin y)\,dy = 0$ is-----.

Select the correct option

Reload M

- $\left(\frac{2}{x}\right)dx + \left(\cos^2 y - \tan y\right)dy = 0$
- $\left(\frac{x}{2}\right)dx + \left(\sec^2 y - \cot y\right)dy = 0$
- $\left(\frac{x}{2}\right)dx + \left(\sec^2 y - \tan y\right)dy = 0$
- $\left(\frac{2}{x}\right)dx + \left(\sec^2 y - \tan y\right)dy = 0$

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Quiz Start Time: 03:58 PM

Question # 26 of 30 (Start time: 04:08:08 PM, 03 January 2021)

If the differential equation: $M(x,y)dx + N(x,y)dy = 0$ is not exact, then after - - - an appropriate function $u(x,y)$, we can from it to be exact.

Select the correct option

Reload

- adding
- subtracting
- multiplying by
- dividing by

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MTH401:Grand Quiz

Quiz Start Time: 03:58 PM, 03 Janu

Question # 27 of 30 (Start time: 04:08:32 PM, 03 January 2021)

Total

If the auxiliary equation, $am^2 + bm + c = 0$, has the complex roots, $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$, where α and $\beta > 0$ are real and $i^2 = -1$. Then the solution of the differential equation, $ay'' + by' + cy = 0$, could be _____.

Select the correct option

Reload Math Eq

- $y = c_1\, e^{(\alpha+i\beta)x}$
- $y = c_2\, e^{(\alpha-i\beta)x}$
- $y = c_1\, e^{(\alpha+i\beta)x} + c_2\, e^{(\alpha-i\beta)x}$
- $y = c_1\, e^{(\alpha+i\beta)x} - c_2\, e^{(\alpha-i\beta)x}$

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MTH401:Grand Quiz

Quiz Start Time: 03:58

Question # 28 of 30 (Start time: 04:08:52 PM, 03 January 2021)

$If\ d\left[\ln(xy)\right] = 0\ is\ anequivalent\ form\ of\ the\ differential\ equation : xdy + ydx = 0, then\ its\ solutionis - --$

Select the correct option

Relo

- ○ $y = c + x$
- ○ $y = c - x$
- ○ $y = cx$
- ◉ $y = \frac{c}{x}$

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Quiz Start Ti

Question # 29 of 30 (Start time: 04:09:11 PM, 03 January 2021)

The solution of the auxiliary equation determines the _____ of the differential equation.

Select the correct option

- order
- degree
- solutions
- linearity

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MTH401:Grand Quiz

Quiz Start Time: 03:5

Question # 30 of 30 (Start time: 04:09:31 PM, 03 January 2021)

For the non - exact differential equation

$$ydx - (y - 3x - 3)\,dy = 0, if \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{2}{y}$$

Select the correct option

Re

○	$\frac{-1}{y^2}$
○	$\frac{1}{y^2}$
○	$-y^2$
◉	y^2

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MTH401:Grand Quiz

Quiz Start Time

Question # 1 of 30 (Start time: 09:54:36 PM, 03 January 2021)

Which of following is an example of Homogeneous function?

Select the correct option

- $f(x,y)=\sin\left(\frac{1}{x}\right)$
- $f(x,y)=\sin\left(\frac{x}{y}\right)$
- $f(x,y)=\sin x$
- $f(x,y)=\sin xy$

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Quiz Start Time: 09:54 PM, (

Question # 2 of 30 (Start time: 09:55:15 PM, 03 January 2021)

For a second order linear ODE, if y_1 and y_2 are its solutions, and further $W(y_1, y_2) = 4$, $W_2 = -\frac{1}{5}$, then $u'_2 =$ _____.

Select the correct option

Reload M

- 4
- -20
- $-\frac{1}{5}$
- $-\frac{1}{20}$

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Quiz Start Time: 09:

Question # 3 of 30 (Start time: 09:55:41 PM, 03 January 2021)

An annihilator operator L for the function, $f(x) = e^{3x}$, is $L =$ _____.

Select the correct option

Re

- [x] $D - 3$
- [ ] $D - e^{3x}$
- [ ] $3e^{3x}$
- [ ] e^{3x}

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Question # 4 of 30 (Start time: 09:56:11 PM, 03 January 2021)

$\frac{d^2y}{dx^2} + 5\left(\frac{dy}{dx}\right)^3 - 3y = e^{\sin x}$ is an example of --------- differential equation.

Select the correct option

○	ordinary linear
◉	ordinary non-linear
○	partial linear
○	partial non-linear

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Question # 5 of 30 (Start time: 09:56:39 PM, 03 January 2021)

What is annihilator operator of the function

$$g(x) = 8e^{3x}$$

?

Select the correct option

○	$(D+3)$
◉	$(D-3)$
○	$(D+3)^2$
○	$(D-3)^2$

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Quiz Start Time: 09:5

Question # 6 of 30 (Start time: 09:57:05 PM, 03 January 2021)

The annihilator operator for the function, $g(x) = 2e^{-x}$, is _____.

Select the correct option

Rel

- D
- D^2
- $D + 1$
- $D + 2$

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Question # 7 of 30 (Start time: 09:57:35 PM, 03 January 2021)

The differential operator D transforms a _____ function into another function.

Select the correct option

- integrable
- differentiable
- continuous
- discontinuous

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Quiz Start

Question # 8 of 30 (Start time: 09:58:04 PM, 03 January 2021)

If we substitute

$$u = \frac{y}{x}$$

in differential equation

Select the correct option

- $ue^{u}du = dx$
- $ue^{(-u)}du = dx$
- $ue^{u}du = dy$
- $ue^{(-u)}du = dy$

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Quiz Start Time: 09:54 PM

Question # 9 of 30 (Start time: 09:58:31 PM, 03 January 2021)

If $A(t)$ is the amount of a radioactive substance present at time t, the the rate of change of $A(t)$ with respect to t is given by _____.

Select the correct option

Reload

- $dA = dt$
- $A_0\, e^t$
- $\frac{A_0}{2}$
- $\frac{dA}{dt} = kA$; k is a constant of proportionality

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Question # 10 of 30 (Start time: 09:58:57 PM, 03 January 2021)

Which of following would be a general solution of the differential equation: $\frac{dy}{dx} = 4$?

Select the correct option

- [x] $y = 4x + a$
- [ ] $y = ax + 4$
- [ ] $y = 4x + 4$
- [ ] $y = ax + a$

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Quiz Start Time

Question # 11 of 30 (Start time: 09:59:26 PM, 03 January 2021)

If the differential equation: $M(x,y)dx + N(x,y)dy = 0$ is not exact, then after - - - an appropriate function $u(x,y)$, we can from it to be exact.

Select the correct option

- adding
- subtracting
- multiplying by
- dividing by

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MTH401:Grand Quiz

Quiz Start Time:

Question # 12 of 30 (Start time: 09:59:51 PM, 03 January 2021)

In exponential model for the population growth

$$P(t) = P_0 e^{kt} \quad If\ k < 0,\ then\ \lim_{t \to \infty} P(t) =$$

Select the correct option

●	0
○	1
○	∞
○	- ∞

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Question # 13 of 30 (Start time: 10:00:14 PM, 03 January 2021)

Which of the following would be a particular solution of the differential equation: $\frac{dy}{dx} = 4$?

Select the correct option

○	$y = 4x + a$
○	$y = ax + 4$
○	$y = ax + a$
◉	$y = 4x + 4$

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Quiz Start Time: 09

Question # 14 of 30 (Start time: 10:01:42 PM, 03 January 2021)

In exponential model for the population growth

$$P(t) = P_0 e^{kt} \quad If\ k > 0,\ then\ \lim_{t \to \infty} P(t) =$$

Select the correct option

○	0
○	1
●	∞
○	- ∞

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Question # 15 of 30 (Start time: 10:02:30 PM, 03 January 2021)

Separable form of the differential equation: $\frac{dy}{dx} = y - 1$ is ------, where $v = y - 1$.

Select the correct option

- $\frac{dx}{x} = dv$
- $dx = vdv$
- $\frac{dv}{v} = dx$
- $\frac{dv}{v} = \frac{dx}{x}$

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MTH401:Grand Quiz

Quiz Start Time: 09:5

Question # 16 of 30 (Start time: 10:03:04 PM, 03 January 2021)

The superposition principle is a property of linear differential equations and it does not hold in case of _____ differential equations.

Select the correct option

- ○ linear
- ◉ nonlinear

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MTH401:Grand Quiz

Quiz Start Time: 09:54 PM, 03 Ja

Question # 17 of 30 (Start time: 10:03:37 PM, 03 January 2021)

To

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = e^{3x} \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math

○	Ae^{5x}
○	$(Ax + B)e^{5x}$
○	$(Ax^2 + Bx + C)e^{5x}$
●	$Ae^{3x} \cos 4x + Be^{3x} \sin 4x$

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Question # 18 of 30 (Start time: 10:04:10 PM, 03 January 2021)

Classify the following differential equation

$$e^x \frac{dy}{dx} + 2y = 3xy$$

Select the correct option

○	Separable and not linear
○	Linear and not separable
◉	Both separable and linear
○	Neither separable nor linear

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Quiz Start T

Question # 19 of 30 (Start time: 10:04:41 PM, 03 January 2021)

A set of functions

$$\{f_1(x), f_2(x), f_3(x), \ldots\ldots, f_n(x)\}$$

is said to be _____ on an interval I if there exist constants

Select the correct option

- ○ linear dependent
- ● linear independent

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Quiz Start Time

Question # 20 of 30 (Start time: 10:05:05 PM, 03 January 2021)

If

$$a_n(x) = 0$$

in the differential equation

Select the correct option

- unique
- exist
- exist and unique
- none of these

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Quiz Start Time: 09:5

Question # 21 of 30 (Start time: 10:05:33 PM, 03 January 2021)

The auxiliary equation of the differential equation, $y''' + 2y'' - 3y = 0$, is _____.

Select the correct option

Re

- [x] $m^3 + 2m^2 - 3 = 0$
- [ ] $\left(m^3 + 2m^2 - 3\right) e^{mx} = 0$
- [ ] $m^3 + 2m^2 = 0$
- [ ] $m^3 - 3 = 0$

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MTH401:Grand Quiz

Quiz Start Time: 09:54 PM, 03 January 2

Question # 22 of 30 (Start time: 10:05:56 PM, 03 January 2021)

Total Mar

For the nonexact differential equation $\left(x^2y - 2xy^2\right)dx - \left(x^3 - 3x^2y\right)dy = 0, if M(x,y) + yN(x,y) = x^2y^2 \neq 0, then\ the\ integrating\ factor\ is - --$

Select the correct option

Reload Math Equatio

- ● $\frac{1}{x^2y^2}$
- ○ $-\frac{1}{x^2y^2}$
- ○ $e^{x^2y^2}$
- ○ $e^{-x^2y^2}$

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MTH401:Grand Quiz

Quiz Start Time: 09:54 PM

Question # 23 of 30 (Start time: 10:06:51 PM, 03 January 2021)

How are frequency and period related in simple harmonic motion?

Select the correct option

- They are directly related
- Their sum is constant
- They are inversely related
- None of the above

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MTH401:Grand Quiz

Quiz Start Time: 09:54 PM, 03 January 2021

Question # 24 of 30 (Start time: 10:07:21 PM, 03 January 2021)

Total Marks: 1

$For\ the\ non-exact\ differential\ equation\ M(x,y)dx+N(x,y)dy=0, if \frac{\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}}{N}\$\ is\ a\ function\ of\ x, then\ the\ integrating\ factor\ is\ a---$

Select the correct option

Reload Math Equations

- [x] function of x also
- [ ] function of y
- [ ] multi - variable function of both x and y
- [ ] constant

Click to Save Answer & Move to Next Question

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MTH401:Grand Quiz

Quiz Start Time: 09

Question # 25 of 30 (Start time: 10:08:09 PM, 03 January 2021)

Separable form $f(y)dy = g(x)dx$, of the differential equation:$y - x\frac{dy}{dx} = a\left(y^2 + \frac{dy}{dx}\right)$ is-----.

Select the correct option

- $\frac{1}{y(1+ay)}dy = \frac{1}{x+a}dx$
- $\frac{1}{y(1-ay)}dy = \frac{1}{x+a}dx$
- $\frac{1}{y(1+ay)}dy = \frac{1}{x-a}dx$
- $\frac{1}{y(1-ay)}dy = \frac{1}{x-a}dx$

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MTH401:Grand Quiz

Quiz Start Time: 09:54 PM, 0

Question # 26 of 30 (Start time: 10:08:29 PM, 03 January 2021)

If we want to solve an initial value problem then we need to to apply the initial conditions to the _____ solution and not to the complementary function alone.

Select the correct option

- complementary
- particular
- general
- exact

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MTH401:Grand Quiz

Question # 27 of 30 (Start time: 10:08:52 PM, 03 January 2021)

The differential equation

$$\frac{d^3y}{dx^3} + y^2 = x^4$$

Select the correct option

○	Non-homogeneous and linear
○	Homogeneous and non linear
◉	Non homogeneous and non-linear
○	None of these

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MTH401:Grand Quiz

Quiz Start Ti

Question # 28 of 30 (Start time: 10:09:12 PM, 03 January 2021)

$$The\ differential\ equation\ \frac{dy}{dx} = \frac{y}{x}\ is\ ---.$$

Select the correct option

○	Exact
●	Non exact
○	Non-linear
○	Inseparable

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MTH401:Grand Quiz

Quiz Start Time: 09:5

Question # 29 of 30 (Start time: 10:09:36 PM, 03 January 2021)

Which of the following is an equivalent form of the exact differential equation:
$ydx + xdy = 0$?

Select the correct option

Rel

- $d(\frac{x}{y}) = 0$
- $d\left(\frac{y}{x}\right) = 0$
- $d(xy) = 0$
- $d(x + y) = 0$

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MTH401:Grand Quiz

Quiz Start Time: 09:54 PM

Question # 30 of 30 (Start time: 10:09:56 PM, 03 January 2021)

Which of the following is first order linear equation in unknown variable x?

Select the correct option

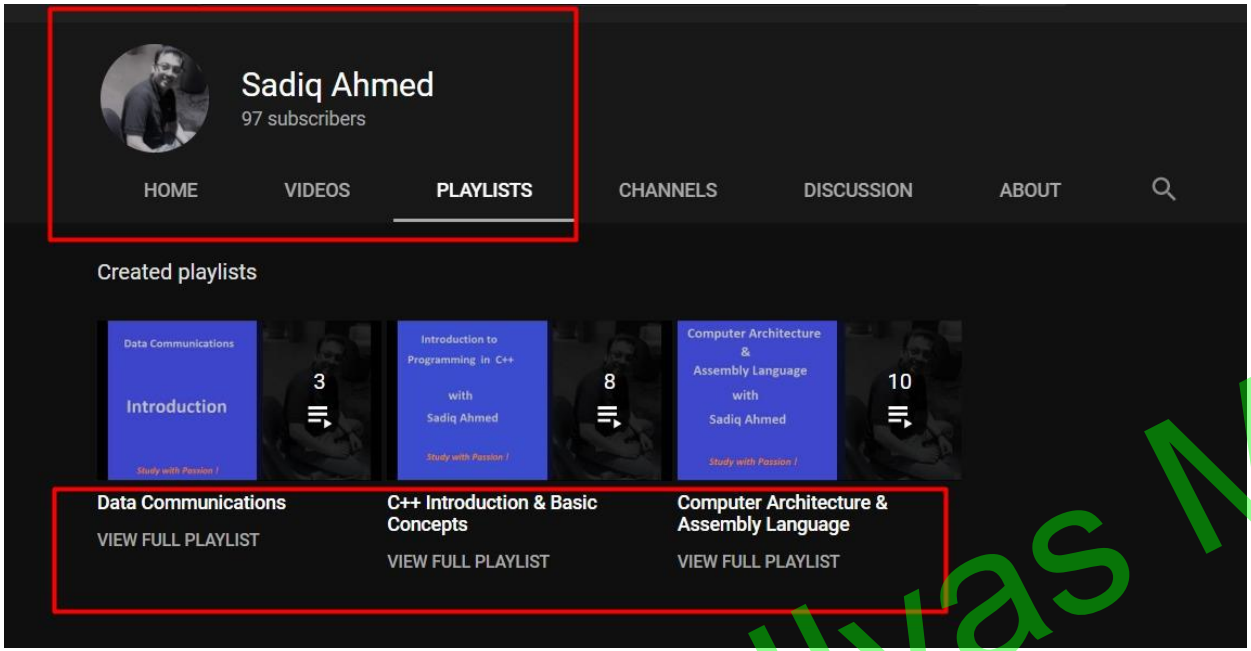
Reload

- $x\frac{dy}{dx} + (\sin x)\, y = \cos x$
- $y\frac{dx}{dy} + (\sin y)\, x = \cos y$
- $y\frac{dx}{dy} + (\sin y)\, x = \cos x$
- $y\frac{dx}{dy} + (\sin x)\, x = \cos y$

Click to Save Answer & Move to

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3) Recorded Videos
4) Topic wise Notes
5) Short QA Solved
6) MCQs Solved
7) Solved Past and Current paper
8) Solved Graded Activities
(Assignment Quiz GDB)

Activity
Detailed
Information

Project Enrolment Complete training with
Live classes on Zoom
1) SRS (Software Requirements Specification)
2) DD (Design Document)
3) Test phase
a) Test phase viva (Live Classes for viva preparation)
4) Final Deliverable
a) Final coding
b) Final presentation
c) Final report
d) Pre final viva (Coding Viva)
e) Final viva (Overall Presentation)

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